

Ch6 Functions Part 1 後測

建議作答時間：20 分鐘

姓名：_____ 學號：_____

Learning Outcome Test

T1

閱讀下列程式碼：

```
function applyDiscount(price) {  
    return price * 0.9;  
}  
  
const selectedRule = applyDiscount;  
const finalPrice = applyDiscount(1000);
```

`selectedRule` 目前代表什麼？

- A. 一個函數物件
- B. `900` 這個執行後的結果
- C. `1000` 這個原始價格
- D. 一個布林值

T2

下列哪一段程式碼在第一行呼叫時可以成功執行？

A.

```
showTotal(500);  
  
function showTotal(total) {  
    return total + 60;  
}
```

B.

```
showTotal(500);  
  
const showTotal = function(total) {  
    return total + 60;  
};
```

C.

```
showTotal(500);  
  
var showTotal = function(total) {  
    return total + 60;  
};
```

D.

```
showTotal(500);

let showTotal = (total) => total + 60;
```

T3

閱讀下列程式碼：

```
function calculateOrder(productTotal, shippingFee, discount) {
  return productTotal + shippingFee - discount;
}

const result = calculateOrder(1200, 80);
```

T3-1

請填入本次呼叫時各 parameter 對應到的值。

parameter	本次呼叫時的值
productTotal	_____
shippingFee	_____
discount	_____

T3-2

`result` 的值或狀態是什麼？

答案：_____

T4

閱讀下列函數：

```
function calculateOrderTotal(productTotal, shippingFee = 60, discount = 0) {
  return productTotal + shippingFee - discount;
}
```

現在要計算一筆商品總額為 1500、運費使用一般設定、折扣為 100 的訂單。下列哪一個呼叫形式最合適？

- A. `calculateOrderTotal(1500)`
- B. `calculateOrderTotal(1500, 60, 100)`
- C. `calculateOrderTotal(1500, 100)`
- D. `calculateOrderTotal(1500, undefined, 100)`

T5

T5-1

某同學想寫一個函數，用來接收一個會員等級與多個評分。請判斷下列函數參數列是否合法，只需在「判斷」欄填入「合法」或「不合法」。

候選參數列	判斷
<code>(level, ...scores)</code>	_____
<code>(...scores, level)</code>	_____
<code>(level, ...scores, ...bonuses)</code>	_____

T5-2

閱讀下列程式碼：

```
function collectScores(level, ...scores) {
  return scores;
}

const values = collectScores("gold", 5, 4, 3);
```

`values` 的內容是什麼？

答案：_____

T6

促銷頁載入時只需要初始化一次活動標籤。比較下列幾種寫法，哪一種較不容易讓臨時計算用的名稱留在外層被後續程式誤用？

A.

```
const campaignName = "Spring Sale";
const badgeText = `Sale: ${campaignName}`;
console.log(badgeText);
```

B.

```
(function() {
  const campaignName = "Spring Sale";
  const badgeText = `Sale: ${campaignName}`;
  console.log(badgeText);
})();
```

C.

```
const campaignName = "Spring Sale";
function badgeText() {
  console.log(`Sale: ${campaignName}`);
}
```

D.

```
let campaignName = "Spring Sale";
let badgeText = "Sale";
console.log(badgeText);
```

T7

閱讀下列程式碼：

```
function calculateShipping(weight, rate) {  
  return weight * rate;  
}  
  
function applyRule(rule, ...args) {  
  return rule(...args);  
}
```

T7-1

在下列呼叫中，哪一個傳入 `applyRule` 第一個位置的是函數物件？

- A. `applyRule(calculateShipping, 8, 30)`
- B. `applyRule(calculateShipping(8, 30))`
- C. `applyRule(240, 8, 30)`
- D. `applyRule("calculateShipping", 8, 30)`

T7-2

接續 T7-1 的程式碼，閱讀下列呼叫：

```
const result = applyRule(calculateShipping, 8, 30);
```

`calculateShipping` 實際收到的兩個引數依序是什麼？

答案：_____，_____

T8

T8-1

訂單清單中每筆訂單都有 `total`。現在只需要把每筆訂單的總額取出，作為簡短 callback 使用。下列哪一個寫法最適合放入 `map`？

- A. `function order(order.total) { return total; }`
- B. `order => order.total`
- C. `(order.total) => order`
- D. `order => { order.total }`

T8-2

只補上 `map` 裡面的 callback 部分，讓 `totals` 取得每筆訂單的總額。

```
const orders = [  
  { id: "A01", total: 980 },  
  { id: "A02", total: 1250 }  
];  
  
const totals = orders.map(_____);
```

T9

閱讀下列程式碼與文件註解：

```
function calculateGoldPoints(amount, multiplier) {
  return amount * multiplier + 100;
}

function calculateSilverPoints(amount, multiplier) {
  return amount * multiplier + 100;
}

/**
 * Calculates member points.
 * @param {number} amount
 * @param {number} multiplier
 */
function calculateMemberPoints(amount, multiplier) {
  return amount * multiplier + 100;
}
```

T9-1

這段程式碼最主要的函數設計問題是什麼？

- A. 兩個函數中出現相同的點數計算邏輯
- B. 函數名稱都太短，無法看出用途
- C. 應該改用 `console.log` 才能確認結果
- D. 會員點數計算應改成字串處理

T9-2

請判斷上方 JSDoc 目前缺少哪一類資訊，只需填入一個代碼。

P = 函數目的 A = 參數 R = 回傳值 N = 沒有缺少

答案：_____

T10

原本程式碼如下：

```
function calculateCouponTotal(price, quantity) {
  return price * quantity + 30;
}

const mayTotal = calculateCouponTotal(200, 4);
const leoTotal = calculateCouponTotal(350, 2);
```

T10-1

若要把服務費改成可省略的預設設定，下列哪一個改寫仍能保留原本兩次呼叫的結果？

A.

```
function calculateCouponTotal(price, quantity, serviceFee = 30) {
  return price * quantity + serviceFee;
}

const mayTotal = calculateCouponTotal(200, 4);
const leoTotal = calculateCouponTotal(350, 2);
```

B.

```
function calculateCouponTotal(price, quantity, serviceFee = 0) {  
  return price * quantity + serviceFee;  
}  
  
const mayTotal = calculateCouponTotal(200, 4);  
const leoTotal = calculateCouponTotal(350, 2);
```

C.

```
function calculateCouponTotal(price, serviceFee = 30, quantity) {  
  return price * quantity + serviceFee;  
}  
  
const mayTotal = calculateCouponTotal(200, 4);  
const leoTotal = calculateCouponTotal(350, 2);
```

D.

```
function calculateCouponTotal(price, quantity, serviceFee = 30) {  
  return price + quantity + serviceFee;  
}  
  
const mayTotal = calculateCouponTotal(200, 4);  
const leoTotal = calculateCouponTotal(350, 2);
```

T10-2

請把下列片段排序，完成一個保留原本兩次呼叫結果的版本。答案只需寫片段代碼順序，例如 `A-B-C-D-E`。

A.

```
const leoTotal = calculateCouponTotal(350, 2);
```

B.

```
function calculateCouponTotal(price, quantity, serviceFee = 30) {
```

C.

```
}
```

D.

```
const mayTotal = calculateCouponTotal(200, 4);
```

E.

```
  return price * quantity + serviceFee;
```

F.

```
return price + quantity + serviceFee;
```

G.

```
const mayTotal = calculateCouponTotal(200);
```

答案：_____